

Chapter 1 Kinematics

Kinematics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 1 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

1.1 Kinematics in practice

Kinematics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 2 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

1.2 Kinematics in practice

Kinematics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 3 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

1.3 Kinematics in practice

Kinematics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 4 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

1.4 Kinematics in practice

Kinematics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 5 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

1.5 Kinematics in practice

Kinematics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 6 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

1.6 Kinematics in practice

Kinematics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 7 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

1.7 Kinematics in practice

Kinematics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 8 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

1.8 Kinematics in practice

Kinematics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 9 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

1.9 Kinematics in practice

Kinematics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 10 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 2 Forces

Forces rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 11 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

2.1 Forces in practice

Forces rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 12 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

2.2 Forces in practice

Forces rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 13 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

2.3 Forces in practice

Forces rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 14 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

2.4 Forces in practice

Forces rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 15 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

2.5 Forces in practice

Forces rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 16 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

2.6 Forces in practice

Forces rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 17 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

2.7 Forces in practice

Forces rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 18 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

2.8 Forces in practice

Forces rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 19 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

2.9 Forces in practice

Forces rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 20 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 3 Energy

Energy rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 21 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

3.1 Energy in practice

Energy rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 22 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

3.2 Energy in practice

Energy rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 23 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

3.3 Energy in practice

Energy rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 24 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

3.4 Energy in practice

Energy rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 25 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

3.5 Energy in practice

Energy rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 26 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

3.6 Energy in practice

Energy rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 27 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

3.7 Energy in practice

Energy rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 28 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

3.8 Energy in practice

Energy rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 29 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

3.9 Energy in practice

Energy rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 30 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 4 Momentum

Momentum rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 31 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

4.1 Momentum in practice

Momentum rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 32 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

4.2 Momentum in practice

Momentum rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 33 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

4.3 Momentum in practice

Momentum rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 34 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

4.4 Momentum in practice

Momentum rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 35 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

4.5 Momentum in practice

Momentum rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 36 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

4.6 Momentum in practice

Momentum rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 37 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

4.7 Momentum in practice

Momentum rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 38 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

4.8 Momentum in practice

Momentum rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 39 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

4.9 Momentum in practice

Momentum rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 40 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 5 Rotation

Rotation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 41 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

5.1 Rotation in practice

Rotation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 42 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

5.2 Rotation in practice

Rotation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 43 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

5.3 Rotation in practice

Rotation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 44 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

5.4 Rotation in practice

Rotation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 45 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

5.5 Rotation in practice

Rotation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 46 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

5.6 Rotation in practice

Rotation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 47 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

5.7 Rotation in practice

Rotation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 48 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

5.8 Rotation in practice

Rotation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 49 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

5.9 Rotation in practice

Rotation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 50 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 6 Gravitation

Gravitation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 51 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

6.1 Gravitation in practice

Gravitation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 52 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

6.2 Gravitation in practice

Gravitation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 53 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

6.3 Gravitation in practice

Gravitation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 54 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

6.4 Gravitation in practice

Gravitation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 55 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

6.5 Gravitation in practice

Gravitation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 56 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

6.6 Gravitation in practice

Gravitation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 57 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

6.7 Gravitation in practice

Gravitation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 58 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

6.8 Gravitation in practice

Gravitation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 59 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

6.9 Gravitation in practice

Gravitation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 60 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 7 Oscillations

Oscillations rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 61 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

7.1 Oscillations in practice

Oscillations rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 62 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

7.2 Oscillations in practice

Oscillations rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 63 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

7.3 Oscillations in practice

Oscillations rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 64 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

7.4 Oscillations in practice

Oscillations rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 65 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

7.5 Oscillations in practice

Oscillations rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 66 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

7.6 Oscillations in practice

Oscillations rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 67 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

7.7 Oscillations in practice

Oscillations rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 68 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

7.8 Oscillations in practice

Oscillations rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 69 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

7.9 Oscillations in practice

Oscillations rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 70 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 8 Waves

Waves rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 71 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

8.1 Waves in practice

Waves rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 72 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

8.2 Waves in practice

Waves rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 73 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

8.3 Waves in practice

Waves rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 74 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

8.4 Waves in practice

Waves rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 75 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

8.5 Waves in practice

Waves rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 76 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

8.6 Waves in practice

Waves rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 77 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

8.7 Waves in practice

Waves rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 78 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

8.8 Waves in practice

Waves rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 79 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

8.9 Waves in practice

Waves rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 80 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 9 Fluids

Fluids rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 81 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

9.1 Fluids in practice

Fluids rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 82 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

9.2 Fluids in practice

Fluids rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 83 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

9.3 Fluids in practice

Fluids rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 84 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

9.4 Fluids in practice

Fluids rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 85 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

9.5 Fluids in practice

Fluids rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 86 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

9.6 Fluids in practice

Fluids rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 87 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

9.7 Fluids in practice

Fluids rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 88 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

9.8 Fluids in practice

Fluids rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 89 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

9.9 Fluids in practice

Fluids rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 90 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 10 Thermodynamics

Thermodynamics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 91 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

10.1 Thermodynamics in practice

Thermodynamics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 92 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

10.2 Thermodynamics in practice

Thermodynamics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 93 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

10.3 Thermodynamics in practice

Thermodynamics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 94 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

10.4 Thermodynamics in practice

Thermodynamics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 95 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

10.5 Thermodynamics in practice

Thermodynamics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 96 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

10.6 Thermodynamics in practice

Thermodynamics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 97 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

10.7 Thermodynamics in practice

Thermodynamics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 98 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

10.8 Thermodynamics in practice

Thermodynamics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 99 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

10.9 Thermodynamics in practice

Thermodynamics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 100 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 11 Electrostatics

Electrostatics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 101 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

11.1 Electrostatics in practice

Electrostatics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 102 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

11.2 Electrostatics in practice

Electrostatics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 103 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

11.3 Electrostatics in practice

Electrostatics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 104 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

11.4 Electrostatics in practice

Electrostatics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 105 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

11.5 Electrostatics in practice

Electrostatics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 106 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

11.6 Electrostatics in practice

Electrostatics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 107 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

11.7 Electrostatics in practice

Electrostatics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 108 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

11.8 Electrostatics in practice

Electrostatics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 109 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

11.9 Electrostatics in practice

Electrostatics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 110 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 12 Circuits

Circuits rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 111 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

12.1 Circuits in practice

Circuits rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 112 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

12.2 Circuits in practice

Circuits rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 113 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

12.3 Circuits in practice

Circuits rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 114 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

12.4 Circuits in practice

Circuits rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 115 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

12.5 Circuits in practice

Circuits rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 116 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

12.6 Circuits in practice

Circuits rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 117 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

12.7 Circuits in practice

Circuits rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 118 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

12.8 Circuits in practice

Circuits rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 119 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

12.9 Circuits in practice

Circuits rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 120 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 13 Magnetism

Magnetism rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 121 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

13.1 Magnetism in practice

Magnetism rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 122 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

13.2 Magnetism in practice

Magnetism rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 123 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

13.3 Magnetism in practice

Magnetism rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 124 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

13.4 Magnetism in practice

Magnetism rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 125 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

13.5 Magnetism in practice

Magnetism rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 126 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

13.6 Magnetism in practice

Magnetism rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 127 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

13.7 Magnetism in practice

Magnetism rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 128 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

13.8 Magnetism in practice

Magnetism rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 129 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

13.9 Magnetism in practice

Magnetism rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 130 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 14 Induction

Induction rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 131 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

14.1 Induction in practice

Induction rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 132 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

14.2 Induction in practice

Induction rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 133 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

14.3 Induction in practice

Induction rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 134 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

14.4 Induction in practice

Induction rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 135 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

14.5 Induction in practice

Induction rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 136 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

14.6 Induction in practice

Induction rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 137 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

14.7 Induction in practice

Induction rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 138 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

14.8 Induction in practice

Induction rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 139 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

14.9 Induction in practice

Induction rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 140 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 15 Optics

Optics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 141 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

15.1 Optics in practice

Optics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 142 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

15.2 Optics in practice

Optics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 143 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

15.3 Optics in practice

Optics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 144 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

15.4 Optics in practice

Optics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 145 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

15.5 Optics in practice

Optics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 146 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

15.6 Optics in practice

Optics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 147 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

15.7 Optics in practice

Optics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 148 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

15.8 Optics in practice

Optics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 149 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

15.9 Optics in practice

Optics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 150 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 16 Relativity

Relativity rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 151 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

16.1 Relativity in practice

Relativity rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 152 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

16.2 Relativity in practice

Relativity rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 153 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

16.3 Relativity in practice

Relativity rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 154 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

16.4 Relativity in practice

Relativity rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 155 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

16.5 Relativity in practice

Relativity rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 156 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

16.6 Relativity in practice

Relativity rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 157 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

16.7 Relativity in practice

Relativity rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 158 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

16.8 Relativity in practice

Relativity rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 159 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

16.9 Relativity in practice

Relativity rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 160 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 17 Kinematics

Kinematics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 161 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

17.1 Kinematics in practice

Kinematics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 162 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

17.2 Kinematics in practice

Kinematics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 163 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

17.3 Kinematics in practice

Kinematics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 164 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

17.4 Kinematics in practice

Kinematics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 165 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

17.5 Kinematics in practice

Kinematics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 166 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

17.6 Kinematics in practice

Kinematics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 167 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

17.7 Kinematics in practice

Kinematics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 168 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

17.8 Kinematics in practice

Kinematics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 169 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

17.9 Kinematics in practice

Kinematics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 170 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 18 Forces

Forces rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 171 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

18.1 Forces in practice

Forces rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 172 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

18.2 Forces in practice

Forces rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 173 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

18.3 Forces in practice

Forces rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 174 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

18.4 Forces in practice

Forces rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 175 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

18.5 Forces in practice

Forces rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 176 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

18.6 Forces in practice

Forces rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 177 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

18.7 Forces in practice

Forces rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 178 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

18.8 Forces in practice

Forces rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 179 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

18.9 Forces in practice

Forces rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 180 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 19 Energy

Energy rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 181 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

19.1 Energy in practice

Energy rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 182 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

19.2 Energy in practice

Energy rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 183 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

19.3 Energy in practice

Energy rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 184 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

19.4 Energy in practice

Energy rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 185 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

19.5 Energy in practice

Energy rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 186 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

19.6 Energy in practice

Energy rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 187 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

19.7 Energy in practice

Energy rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 188 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

19.8 Energy in practice

Energy rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 189 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

19.9 Energy in practice

Energy rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 190 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 20 Momentum

Momentum rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 191 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

20.1 Momentum in practice

Momentum rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 192 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

20.2 Momentum in practice

Momentum rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 193 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

20.3 Momentum in practice

Momentum rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 194 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

20.4 Momentum in practice

Momentum rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 195 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

20.5 Momentum in practice

Momentum rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 196 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

20.6 Momentum in practice

Momentum rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 197 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

20.7 Momentum in practice

Momentum rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 198 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

20.8 Momentum in practice

Momentum rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 199 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

20.9 Momentum in practice

Momentum rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 200 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 21 Rotation

Rotation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 201 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

21.1 Rotation in practice

Rotation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 202 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

21.2 Rotation in practice

Rotation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 203 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

21.3 Rotation in practice

Rotation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 204 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

21.4 Rotation in practice

Rotation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 205 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

21.5 Rotation in practice

Rotation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 206 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

21.6 Rotation in practice

Rotation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 207 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

21.7 Rotation in practice

Rotation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 208 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

21.8 Rotation in practice

Rotation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 209 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

21.9 Rotation in practice

Rotation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 210 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 22 Gravitation

Gravitation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 211 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

22.1 Gravitation in practice

Gravitation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 212 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

22.2 Gravitation in practice

Gravitation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 213 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

22.3 Gravitation in practice

Gravitation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 214 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

22.4 Gravitation in practice

Gravitation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 215 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

22.5 Gravitation in practice

Gravitation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 216 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

22.6 Gravitation in practice

Gravitation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 217 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

22.7 Gravitation in practice

Gravitation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 218 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

22.8 Gravitation in practice

Gravitation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 219 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

22.9 Gravitation in practice

Gravitation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 220 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 23 Oscillations

Oscillations rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 221 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

23.1 Oscillations in practice

Oscillations rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 222 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

23.2 Oscillations in practice

Oscillations rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 223 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

23.3 Oscillations in practice

Oscillations rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 224 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

23.4 Oscillations in practice

Oscillations rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 225 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

23.5 Oscillations in practice

Oscillations rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 226 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

23.6 Oscillations in practice

Oscillations rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 227 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

23.7 Oscillations in practice

Oscillations rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 228 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

23.8 Oscillations in practice

Oscillations rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 229 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

23.9 Oscillations in practice

Oscillations rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 230 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 24 Waves

Waves rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 231 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

24.1 Waves in practice

Waves rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 232 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

24.2 Waves in practice

Waves rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 233 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

24.3 Waves in practice

Waves rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 234 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

24.4 Waves in practice

Waves rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 235 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

24.5 Waves in practice

Waves rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 236 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

24.6 Waves in practice

Waves rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 237 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

24.7 Waves in practice

Waves rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 238 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

24.8 Waves in practice

Waves rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 239 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

24.9 Waves in practice

Waves rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 240 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 25 Fluids

Fluids rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 241 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

25.1 Fluids in practice

Fluids rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 242 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

25.2 Fluids in practice

Fluids rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 243 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

25.3 Fluids in practice

Fluids rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 244 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

25.4 Fluids in practice

Fluids rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 245 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

25.5 Fluids in practice

Fluids rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 246 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

25.6 Fluids in practice

Fluids rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 247 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

25.7 Fluids in practice

Fluids rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 248 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

25.8 Fluids in practice

Fluids rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 249 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

25.9 Fluids in practice

Fluids rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 250 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 26 Thermodynamics

Thermodynamics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 251 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

26.1 Thermodynamics in practice

Thermodynamics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 252 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

26.2 Thermodynamics in practice

Thermodynamics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 253 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

26.3 Thermodynamics in practice

Thermodynamics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 254 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

26.4 Thermodynamics in practice

Thermodynamics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 255 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

26.5 Thermodynamics in practice

Thermodynamics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 256 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

26.6 Thermodynamics in practice

Thermodynamics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 257 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

26.7 Thermodynamics in practice

Thermodynamics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 258 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

26.8 Thermodynamics in practice

Thermodynamics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 259 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

26.9 Thermodynamics in practice

Thermodynamics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 260 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 27 Electrostatics

Electrostatics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 261 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

27.1 Electrostatics in practice

Electrostatics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 262 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

27.2 Electrostatics in practice

Electrostatics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 263 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

27.3 Electrostatics in practice

Electrostatics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 264 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

27.4 Electrostatics in practice

Electrostatics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 265 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

27.5 Electrostatics in practice

Electrostatics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 266 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

27.6 Electrostatics in practice

Electrostatics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 267 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

27.7 Electrostatics in practice

Electrostatics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 268 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

27.8 Electrostatics in practice

Electrostatics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 269 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

27.9 Electrostatics in practice

Electrostatics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 270 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 28 Circuits

Circuits rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 271 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

28.1 Circuits in practice

Circuits rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 272 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

28.2 Circuits in practice

Circuits rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 273 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

28.3 Circuits in practice

Circuits rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 274 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

28.4 Circuits in practice

Circuits rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 275 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

28.5 Circuits in practice

Circuits rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 276 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

28.6 Circuits in practice

Circuits rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 277 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

28.7 Circuits in practice

Circuits rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 278 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

28.8 Circuits in practice

Circuits rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 279 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

28.9 Circuits in practice

Circuits rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 280 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 29 Magnetism

Magnetism rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 281 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

29.1 Magnetism in practice

Magnetism rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 282 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

29.2 Magnetism in practice

Magnetism rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 283 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

29.3 Magnetism in practice

Magnetism rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 284 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

29.4 Magnetism in practice

Magnetism rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 285 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

29.5 Magnetism in practice

Magnetism rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 286 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

29.6 Magnetism in practice

Magnetism rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 287 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

29.7 Magnetism in practice

Magnetism rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 288 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

29.8 Magnetism in practice

Magnetism rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 289 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

29.9 Magnetism in practice

Magnetism rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 290 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 30 Induction

Induction rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 291 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

30.1 Induction in practice

Induction rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 292 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

30.2 Induction in practice

Induction rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 293 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

30.3 Induction in practice

Induction rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 294 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

30.4 Induction in practice

Induction rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 295 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

30.5 Induction in practice

Induction rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 296 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

30.6 Induction in practice

Induction rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 297 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

30.7 Induction in practice

Induction rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 298 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

30.8 Induction in practice

Induction rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 299 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

30.9 Induction in practice

Induction rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 300 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 31 Optics

Optics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 301 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

31.1 Optics in practice

Optics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 302 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

31.2 Optics in practice

Optics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 303 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

31.3 Optics in practice

Optics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 304 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

31.4 Optics in practice

Optics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 305 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

31.5 Optics in practice

Optics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 306 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

31.6 Optics in practice

Optics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 307 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

31.7 Optics in practice

Optics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 308 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

31.8 Optics in practice

Optics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 309 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

31.9 Optics in practice

Optics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 310 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 32 Relativity

Relativity rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 311 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

32.1 Relativity in practice

Relativity rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 312 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

32.2 Relativity in practice

Relativity rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 313 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

32.3 Relativity in practice

Relativity rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 314 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

32.4 Relativity in practice

Relativity rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 315 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

32.5 Relativity in practice

Relativity rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 316 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

32.6 Relativity in practice

Relativity rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 317 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

32.7 Relativity in practice

Relativity rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 318 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

32.8 Relativity in practice

Relativity rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 319 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

32.9 Relativity in practice

Relativity rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 320 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 33 Kinematics

Kinematics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 321 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

33.1 Kinematics in practice

Kinematics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 322 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

33.2 Kinematics in practice

Kinematics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 323 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

33.3 Kinematics in practice

Kinematics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 324 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

33.4 Kinematics in practice

Kinematics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 325 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

33.5 Kinematics in practice

Kinematics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 326 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

33.6 Kinematics in practice

Kinematics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 327 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

33.7 Kinematics in practice

Kinematics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 328 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

33.8 Kinematics in practice

Kinematics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 329 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

33.9 Kinematics in practice

Kinematics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 330 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 34 Forces

Forces rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 331 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

34.1 Forces in practice

Forces rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 332 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

34.2 Forces in practice

Forces rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 333 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

34.3 Forces in practice

Forces rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 334 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

34.4 Forces in practice

Forces rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 335 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

34.5 Forces in practice

Forces rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 336 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

34.6 Forces in practice

Forces rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 337 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

34.7 Forces in practice

Forces rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 338 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

34.8 Forces in practice

Forces rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 339 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

34.9 Forces in practice

Forces rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 340 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 35 Energy

Energy rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 341 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

35.1 Energy in practice

Energy rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 342 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

35.2 Energy in practice

Energy rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 343 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

35.3 Energy in practice

Energy rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 344 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

35.4 Energy in practice

Energy rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 345 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

35.5 Energy in practice

Energy rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 346 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

35.6 Energy in practice

Energy rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 347 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

35.7 Energy in practice

Energy rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 348 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

35.8 Energy in practice

Energy rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 349 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

35.9 Energy in practice

Energy rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 350 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 36 Momentum

Momentum rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 351 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

36.1 Momentum in practice

Momentum rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 352 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

36.2 Momentum in practice

Momentum rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 353 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

36.3 Momentum in practice

Momentum rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 354 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

36.4 Momentum in practice

Momentum rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 355 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

36.5 Momentum in practice

Momentum rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 356 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

36.6 Momentum in practice

Momentum rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 357 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

36.7 Momentum in practice

Momentum rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 358 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

36.8 Momentum in practice

Momentum rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 359 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

36.9 Momentum in practice

Momentum rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 360 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 37 Rotation

Rotation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 361 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

37.1 Rotation in practice

Rotation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 362 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

37.2 Rotation in practice

Rotation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 363 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

37.3 Rotation in practice

Rotation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 364 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

37.4 Rotation in practice

Rotation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 365 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

37.5 Rotation in practice

Rotation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 366 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

37.6 Rotation in practice

Rotation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 367 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

37.7 Rotation in practice

Rotation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 368 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

37.8 Rotation in practice

Rotation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 369 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

37.9 Rotation in practice

Rotation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 370 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 38 Gravitation

Gravitation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 371 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

38.1 Gravitation in practice

Gravitation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 372 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

38.2 Gravitation in practice

Gravitation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 373 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

38.3 Gravitation in practice

Gravitation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 374 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

38.4 Gravitation in practice

Gravitation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 375 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

38.5 Gravitation in practice

Gravitation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 376 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

38.6 Gravitation in practice

Gravitation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 377 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

38.7 Gravitation in practice

Gravitation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 378 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

38.8 Gravitation in practice

Gravitation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 379 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

38.9 Gravitation in practice

Gravitation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 380 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 39 Oscillations

Oscillations rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 381 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

39.1 Oscillations in practice

Oscillations rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 382 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

39.2 Oscillations in practice

Oscillations rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 383 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

39.3 Oscillations in practice

Oscillations rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 384 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

39.4 Oscillations in practice

Oscillations rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 385 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

39.5 Oscillations in practice

Oscillations rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 386 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

39.6 Oscillations in practice

Oscillations rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 387 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

39.7 Oscillations in practice

Oscillations rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 388 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

39.8 Oscillations in practice

Oscillations rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 389 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

39.9 Oscillations in practice

Oscillations rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 390 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 40 Waves

Waves rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 391 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

40.1 Waves in practice

Waves rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 392 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

40.2 Waves in practice

Waves rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 393 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

40.3 Waves in practice

Waves rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 394 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

40.4 Waves in practice

Waves rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 395 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

40.5 Waves in practice

Waves rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 396 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

40.6 Waves in practice

Waves rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 397 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

40.7 Waves in practice

Waves rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 398 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

40.8 Waves in practice

Waves rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 399 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

40.9 Waves in practice

Waves rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 400 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 41 Fluids

Fluids rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 401 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

41.1 Fluids in practice

Fluids rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 402 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

41.2 Fluids in practice

Fluids rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 403 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

41.3 Fluids in practice

Fluids rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 404 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

41.4 Fluids in practice

Fluids rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 405 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

41.5 Fluids in practice

Fluids rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 406 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

41.6 Fluids in practice

Fluids rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 407 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

41.7 Fluids in practice

Fluids rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 408 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

41.8 Fluids in practice

Fluids rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 409 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

41.9 Fluids in practice

Fluids rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 410 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 42 Thermodynamics

Thermodynamics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 411 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

42.1 Thermodynamics in practice

Thermodynamics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 412 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

42.2 Thermodynamics in practice

Thermodynamics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 413 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

42.3 Thermodynamics in practice

Thermodynamics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 414 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

42.4 Thermodynamics in practice

Thermodynamics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 415 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

42.5 Thermodynamics in practice

Thermodynamics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 416 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

42.6 Thermodynamics in practice

Thermodynamics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 417 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

42.7 Thermodynamics in practice

Thermodynamics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 418 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

42.8 Thermodynamics in practice

Thermodynamics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 419 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

42.9 Thermodynamics in practice

Thermodynamics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 420 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 43 Electrostatics

Electrostatics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 421 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

43.1 Electrostatics in practice

Electrostatics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 422 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

43.2 Electrostatics in practice

Electrostatics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 423 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

43.3 Electrostatics in practice

Electrostatics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 424 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

43.4 Electrostatics in practice

Electrostatics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 425 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

43.5 Electrostatics in practice

Electrostatics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 426 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

43.6 Electrostatics in practice

Electrostatics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 427 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

43.7 Electrostatics in practice

Electrostatics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 428 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

43.8 Electrostatics in practice

Electrostatics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 429 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

43.9 Electrostatics in practice

Electrostatics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 430 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 44 Circuits

Circuits rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 431 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

44.1 Circuits in practice

Circuits rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 432 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

44.2 Circuits in practice

Circuits rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 433 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

44.3 Circuits in practice

Circuits rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 434 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

44.4 Circuits in practice

Circuits rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 435 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

44.5 Circuits in practice

Circuits rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 436 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

44.6 Circuits in practice

Circuits rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 437 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

44.7 Circuits in practice

Circuits rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 438 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

44.8 Circuits in practice

Circuits rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 439 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

44.9 Circuits in practice

Circuits rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 440 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 45 Magnetism

Magnetism rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 441 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

45.1 Magnetism in practice

Magnetism rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 442 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

45.2 Magnetism in practice

Magnetism rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 443 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

45.3 Magnetism in practice

Magnetism rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 444 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

45.4 Magnetism in practice

Magnetism rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 445 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

45.5 Magnetism in practice

Magnetism rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 446 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

45.6 Magnetism in practice

Magnetism rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 447 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

45.7 Magnetism in practice

Magnetism rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 448 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

45.8 Magnetism in practice

Magnetism rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 449 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

45.9 Magnetism in practice

Magnetism rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 450 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 46 Induction

Induction rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 451 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

46.1 Induction in practice

Induction rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 452 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

46.2 Induction in practice

Induction rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 453 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

46.3 Induction in practice

Induction rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 454 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

46.4 Induction in practice

Induction rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 455 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

46.5 Induction in practice

Induction rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 456 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

46.6 Induction in practice

Induction rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 457 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

46.7 Induction in practice

Induction rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 458 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

46.8 Induction in practice

Induction rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 459 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

46.9 Induction in practice

Induction rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 460 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 47 Optics

Optics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 461 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

47.1 Optics in practice

Optics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 462 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

47.2 Optics in practice

Optics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 463 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

47.3 Optics in practice

Optics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 464 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

47.4 Optics in practice

Optics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 465 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

47.5 Optics in practice

Optics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 466 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

47.6 Optics in practice

Optics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 467 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

47.7 Optics in practice

Optics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 468 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

47.8 Optics in practice

Optics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 469 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

47.9 Optics in practice

Optics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 470 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 48 Relativity

Relativity rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 471 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

48.1 Relativity in practice

Relativity rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 472 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

48.2 Relativity in practice

Relativity rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 473 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

48.3 Relativity in practice

Relativity rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 474 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

48.4 Relativity in practice

Relativity rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 475 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

48.5 Relativity in practice

Relativity rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 476 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

48.6 Relativity in practice

Relativity rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 477 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

48.7 Relativity in practice

Relativity rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 478 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

48.8 Relativity in practice

Relativity rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 479 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

48.9 Relativity in practice

Relativity rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 480 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 49 Kinematics

Kinematics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 481 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

49.1 Kinematics in practice

Kinematics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 482 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

49.2 Kinematics in practice

Kinematics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 483 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

49.3 Kinematics in practice

Kinematics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 484 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

49.4 Kinematics in practice

Kinematics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 485 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

49.5 Kinematics in practice

Kinematics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 486 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

49.6 Kinematics in practice

Kinematics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 487 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

49.7 Kinematics in practice

Kinematics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 488 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

49.8 Kinematics in practice

Kinematics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 489 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

49.9 Kinematics in practice

Kinematics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 490 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 50 Forces

Forces rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 491 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

50.1 Forces in practice

Forces rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 492 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

50.2 Forces in practice

Forces rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 493 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

50.3 Forces in practice

Forces rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 494 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

50.4 Forces in practice

Forces rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 495 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

50.5 Forces in practice

Forces rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 496 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

50.6 Forces in practice

Forces rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 497 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

50.7 Forces in practice

Forces rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 498 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

50.8 Forces in practice

Forces rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 499 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

50.9 Forces in practice

Forces rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 500 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 51 Energy

Energy rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 501 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

51.1 Energy in practice

Energy rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 502 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

51.2 Energy in practice

Energy rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 503 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

51.3 Energy in practice

Energy rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 504 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

51.4 Energy in practice

Energy rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 505 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

51.5 Energy in practice

Energy rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 506 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

51.6 Energy in practice

Energy rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 507 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

51.7 Energy in practice

Energy rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 508 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

51.8 Energy in practice

Energy rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 509 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

51.9 Energy in practice

Energy rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 510 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 52 Momentum

Momentum rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 511 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

52.1 Momentum in practice

Momentum rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 512 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

52.2 Momentum in practice

Momentum rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 513 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

52.3 Momentum in practice

Momentum rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 514 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

52.4 Momentum in practice

Momentum rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 515 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

52.5 Momentum in practice

Momentum rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 516 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

52.6 Momentum in practice

Momentum rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 517 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

52.7 Momentum in practice

Momentum rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 518 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

52.8 Momentum in practice

Momentum rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 519 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

52.9 Momentum in practice

Momentum rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 520 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 53 Rotation

Rotation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 521 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

53.1 Rotation in practice

Rotation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 522 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

53.2 Rotation in practice

Rotation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 523 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

53.3 Rotation in practice

Rotation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 524 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

53.4 Rotation in practice

Rotation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 525 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

53.5 Rotation in practice

Rotation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 526 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

53.6 Rotation in practice

Rotation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 527 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

53.7 Rotation in practice

Rotation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 528 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

53.8 Rotation in practice

Rotation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 529 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

53.9 Rotation in practice

Rotation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 530 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 54 Gravitation

Gravitation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 531 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

54.1 Gravitation in practice

Gravitation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 532 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

54.2 Gravitation in practice

Gravitation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 533 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

54.3 Gravitation in practice

Gravitation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 534 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

54.4 Gravitation in practice

Gravitation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 535 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

54.5 Gravitation in practice

Gravitation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 536 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

54.6 Gravitation in practice

Gravitation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 537 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

54.7 Gravitation in practice

Gravitation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 538 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

54.8 Gravitation in practice

Gravitation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 539 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

54.9 Gravitation in practice

Gravitation rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 540 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 55 Oscillations

Oscillations rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 541 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

55.1 Oscillations in practice

Oscillations rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 542 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

55.2 Oscillations in practice

Oscillations rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 543 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

55.3 Oscillations in practice

Oscillations rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 544 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

55.4 Oscillations in practice

Oscillations rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 545 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

55.5 Oscillations in practice

Oscillations rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 546 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

55.6 Oscillations in practice

Oscillations rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 547 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

55.7 Oscillations in practice

Oscillations rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 548 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

55.8 Oscillations in practice

Oscillations rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 549 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

55.9 Oscillations in practice

Oscillations rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 550 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 56 Waves

Waves rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 551 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

56.1 Waves in practice

Waves rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 552 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

56.2 Waves in practice

Waves rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 553 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

56.3 Waves in practice

Waves rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 554 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

56.4 Waves in practice

Waves rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 555 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

56.5 Waves in practice

Waves rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 556 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

56.6 Waves in practice

Waves rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 557 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

56.7 Waves in practice

Waves rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 558 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

56.8 Waves in practice

Waves rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 559 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

56.9 Waves in practice

Waves rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 560 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 57 Fluids

Fluids rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 561 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

57.1 Fluids in practice

Fluids rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 562 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

57.2 Fluids in practice

Fluids rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 563 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

57.3 Fluids in practice

Fluids rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 564 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

57.4 Fluids in practice

Fluids rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 565 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

57.5 Fluids in practice

Fluids rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 566 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

57.6 Fluids in practice

Fluids rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 567 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

57.7 Fluids in practice

Fluids rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 568 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

57.8 Fluids in practice

Fluids rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 569 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

57.9 Fluids in practice

Fluids rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 570 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 58 Thermodynamics

Thermodynamics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 571 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

58.1 Thermodynamics in practice

Thermodynamics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 572 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

58.2 Thermodynamics in practice

Thermodynamics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 573 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

58.3 Thermodynamics in practice

Thermodynamics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 574 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

58.4 Thermodynamics in practice

Thermodynamics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 575 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

58.5 Thermodynamics in practice

Thermodynamics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 576 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

58.6 Thermodynamics in practice

Thermodynamics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 577 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

58.7 Thermodynamics in practice

Thermodynamics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 578 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

58.8 Thermodynamics in practice

Thermodynamics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 579 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

58.9 Thermodynamics in practice

Thermodynamics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 580 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 59 Electrostatics

Electrostatics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 581 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

59.1 Electrostatics in practice

Electrostatics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 582 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

59.2 Electrostatics in practice

Electrostatics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 583 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

59.3 Electrostatics in practice

Electrostatics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 584 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

59.4 Electrostatics in practice

Electrostatics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 585 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

59.5 Electrostatics in practice

Electrostatics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 586 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

59.6 Electrostatics in practice

Electrostatics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 587 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

59.7 Electrostatics in practice

Electrostatics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 588 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

59.8 Electrostatics in practice

Electrostatics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 589 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

59.9 Electrostatics in practice

Electrostatics rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 590 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

Chapter 60 Circuits

Circuits rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 591 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

60.1 Circuits in practice

Circuits rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 592 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

60.2 Circuits in practice

Circuits rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 593 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

60.3 Circuits in practice

Circuits rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 594 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

60.4 Circuits in practice

Circuits rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 595 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

60.5 Circuits in practice

Circuits rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 596 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

60.6 Circuits in practice

Circuits rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 597 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

60.7 Circuits in practice

Circuits rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 598 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

60.8 Circuits in practice

Circuits rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 599 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.

60.9 Circuits in practice

Circuits rests on a small set of definitions that the following pages develop in order.

The first result relates the quantities defined above and is proved from the axioms alone.

A worked example on page 600 shows the computation step by step with units carried through.

Common mistakes include dropping a sign, mixing reference frames and forgetting the normal force.

The exercises at the end of the section extend the example to two and three dimensions.

Later chapters reuse this idea, so the reader should be able to reproduce the derivation unaided.